Iloilo Startup Hackathon Challenge Form

A. Department/Office Name

Office of the Provincial Agriculturist (OPAg) – Iloilo Rice Processing Complex (IRPC), Brgy. Amamaros, Pototan, Iloilo

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B. Focal Person (Name and Position)

Aileen Tanalgo – Assistant Plant Manager, Iloilo Rice Processing Complex

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C. Sector/s Affected or Served by the Challenge

- Rice farmers, most of them smallholders tilling less than two hectares
- Farmer cooperatives and irrigators' associations
- Rice retailers, wholesalers, and institutional buyers
- Indigent households served by the Province's ₱20/kg subsidized rice program
- IRPC plant personnel and Municipal Agriculture Offices (MAOs)

D. Challenge Title

A Digital Operating and Traceability Platform for the Iloilo Rice Processing Complex: Filling Idle Capacity, Speeding Up Procurement, and Tracking Rice from Farmer to Buyer

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E. Challenge Statement

The IRPC is the largest integrated rice processing facility in the region, with five mechanical dryers, a 50-ton-per-day mill, and 1,100 tons of storage. Yet it runs far below capacity for much of the year. Farmers and buyers cannot see prices, schedules, or available slots ahead of time, so the plant is overwhelmed during peak harvest and nearly idle in the off-season. Private mills out-compete us on procurement speed and pricing because they decide and respond faster. Because the mills draw heavy power regardless of volume, electricity alone eats 24 to 28 percent of income, and we only see the cost of each run after the monthly books close. Underlying all of this, the plant has no documented operational system: bookings, the logbook, the bag slip, weighing, and batch records live on paper and in people's heads, so the way the plant runs is inconsistent and cannot be monitored in real time. The plant lost money in 2024 and 2025 and turned its first profitable quarter only in early 2026. We need a digital backbone that encodes how the plant operates end to end — one that fills idle capacity, lets us respond to farmers faster, shows the true cost and margin of every drying and milling run as it happens, and traces each batch from the farmer through

drying and milling to the first-hand buyer it is sold to or first delivered to — every person, group, or organization that purchases directly from the IRPC rice mill, including wholesalers, stores, municipal drop-points, and the ₱20/kg program.

F. Current Response or Setup

Farmers and buyers either call the IRPC office to book or come in as walk-ins. Each transaction is first entered in a logbook. On pickup day, a slip records the number of bags gathered; when the palay arrives at the complex it is weighed, and the same form captures the actual weight. The farmer validates that slip, either in person or after being informed by message, and the same validation is repeated after drying. Palay buying prices are posted on a board in the office and published weekly on the IRPC Facebook page, and currently follow the prevailing market price; the plan is to shift to NFA prices once the IRPC operates as a Local Economic Enterprise under the Provincial Government. Income, costs, and electricity use are consolidated into monthly financial statements after the fact. Since early 2026, management has cut overhead through monthly expense caps and leaner casual labor, producing the first back-to-back profitable months, but these gains rely on manual monitoring rather than a system.

G. Desired Outcome or Ideal Scenario

A simple digital platform with two sides. On the farmer and buyer side, users can check the day's palay buying price and service rates, book drying and milling slots, offer palay for sale with variety and estimated volume, and get notified when their turn is near. On the plant side, management sees a live dashboard of the queue, dryer and mill utilization, electricity cost and gross margin per batch, cash position, a harvest-calendar forecast that helps schedule loads to cut idle running and draw clients during off-peak months, and batch-level traceability that follows palay from each farmer through drying and milling to the first-hand buyer — every person, group, or organization that buys directly from or takes first delivery from the IRPC rice mill, including wholesalers, stores, municipal drop-points, and the ₱20/kg program — so the Province can show where its rice comes from and where it goes. Success means higher utilization across the year, faster procurement decisions, lower electricity cost per kilo processed, and steady surpluses that can support the ₱20/kg rice program.

H. Data Sets, Documents, or Insights Available

- Monthly financial statements from August 2024 to March 2026, with income and costs by revenue stream
- Performance Analyses and Continuance Feasibility Study, including a competitor benchmark of private mills
- IRPC Business Plan and five-year financial projections (2027–2031)
- Three years of palay procurement data by municipality (2021–2023)
- List of 20 regular rice buyers with annual volumes (available in anonymized form)

- Plant specifications, process flow, and current service rates for drying and milling
- Provincial and regional rice production data from PSA

I. Conditions or Considerations for a Good Solution

- Mobile-friendly and usable by farmers with basic phones, with an SMS option where data is weak
- Available in Hiligaynon and English, with a very simple interface
- Records must support government accounting and audit requirements and must not replace official receipts
- Compliant with the Data Privacy Act for farmer and buyer information
- Low-cost to run and maintain, with no heavy new hardware required
- May be built in phases, with teams free to deliver a priority module first (for example booking and queueing, real-time batch costing, or farmer-to-buyer traceability) and expand from there
- Able to link with the farmer registry and MAO records where possible

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J. Additional Notes or Clarifications (Optional)

We are open to pilot-testing the solution at the IRPC with farmer cooperatives in Pototan, our largest palay source, during the next harvest season, then scaling the prototype province-wide after feedback. The solution should support the Province's plan to operate the IRPC as a Local Economic Enterprise and, in time, feed into OPAg's MoRProGRes performance dashboard.